

Multiphase Bidirectional DC-DC Converter with Phase shedding for BLDC Driven Electric Vehicle

Ramya S
PG Student
Department of EEE
BMS College of Engineering
Bangalore, India
ramyas.eee19@bmsce.ac.in

Divya S
Assistant Professor
Department of EEE
BMS College of Engineering
Bangalore, India
divyas.eee@bmsce.ac.in

Chandasree Das
Associate Professor
Department of EEE
BMS College of Engineering
Bangalore, India
chandasreedas.eee@bmsce.ac.in

Abstract—Electric Vehicles are seeking more attention in recent days due to pollution concerns and fear of exhausting petroleum products. In battery powered electric vehicle the use of bi-directional converters would support charging of battery during regenerative braking. Use of multiphase converters offer several advantages such as reduced ripple and reduced passive components size and avoid local hotspot creation and improve system efficiency. However, all the phases in multiphase converters would not be required to operate all the time. Hence this paper focuses on operating the number of phases adaptive to load current. Analysis of interleaved soft switching boost converter to minimize the switching losses is also presented. The BLDC motor is considered as a load which makes it a usual choice because of its various advantages.

Keywords—Electric Vehicle, Phase shedding, Multiphase bi-directional converters, soft switching

I. INTRODUCTION

Increased air pollution concerns and a need to find alternatives to Internal Combustion Engine Vehicles (ICEVs) the automobile industries and the researchers are focusing more on Electric Vehicles (EV). The power conversion efficiency of ICEV is observed to be less than that of EV as the power from the engine has to be transferred through various channels before reaching the wheels[1]. There are mainly two types of EV: All electric vehicle(AEV) and Plug in hybrid electric vehicle(PHEV). Type of EV considered in this paper is all Electric Vehicle powered by a battery. In urban driving there would be frequent stop-and-go conditions which require more energy during braking. Hence a regenerative braking would help to extract the surplus Kinetic energy using which the battery can be charged [2]. To support this two-quadrant operation of electric motor in motoring and the braking mode, a bi-directional DC-DC converter is used [3]. Multiphase converters consist of several power stages in parallel with each phase capable of handling particular current range. The pulses given to the multiphase converters are phase shifted because of which the inductor ripple of individual phase cancels and hence it offers reduced ripple, reduced size of passive components such as inductor and capacitors [4,5]. However, in multiphase converters the number of switches is more compared to single phase converters hence contributing to increased switching loss which can be seen dominant during light load conditions. The concept of phase shedding in multi-phase converters can be used to solve this problem so when a particular phase is not required to operate it could be shed, this would help to improve efficiency [6,7]. The motor considered is a BLDC motor because of its various advantages such as less maintenance, less fuel consumption and more power to volume ratio which makes them attractive in the EV applications compared to other motors [8].

The energy stored in the battery is limited which is connected to the load through converter. Hence focusing on

improving the converter efficiency would help to extend battery life to some extent. The efficiency of bi-directional dc-dc converter can be improved by methods like dynamic power management strategy i.e., to turn off the converter modules/phases when not required in multiphase boost converter and also by employing soft switching mechanism converter efficiency could be further improved [9].

The current paper focusses on load adaptive phase shedding for all electric vehicle driven by BLDC motor, along with which regenerative braking is also employed and the system is simulated using MATLAB/Simulink. The paper is organized in this way Section I Introduction, Section II about System Configuration, Section III Design and Specification, Section IV Simulation results and Discussion, Section V Soft switching and Section VI Conclusion and Future scope.

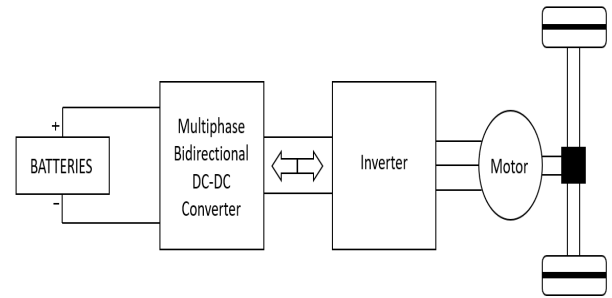


Fig.1. Battery Driven BLDC motor with multi-phase bidirectional DC-DC converter

II. SYSTEM CONFIGURATION

Fig.1 represents the block diagram of the proposed system which consists of battery, multiphase bi-directional dc-dc converter, three phase inverter and BLDC motor which is considered as load. In the motoring mode power flows from battery to the motor through converter and chopper circuit, during regenerative braking power flows in the reverse direction hence recharging the battery. Detailed description of each system component is presented below.

A. Battery

The battery in an Electric vehicle serves as a fuel which powers the electric motor. Various cells could be placed in series-parallel combination to obtain the required voltage and current rating. The energy that is stored in the battery is always limited and precious hence its selection becomes very crucial. Li-ion batteries are the most preferred among the various other types because of its high power and energy density, long life, low maintenance cost, low self-discharge, and high efficiency. On the other hand, these batteries have to be functioned very cautiously because of insignificant tolerance to offensive charge and discharge cycles[10]. The battery is required to operate in two different modes: acceleration and braking. During acceleration the power is drawn from the battery i.e.,

voltage decreases and the SOC(State-of-charge) of battery decreases, during the braking mode the battery will be charged thereby SOC of the battery increases.

B. Multiphase Bi-directional DC-DC converter

The battery is considered to be in low voltage side and the traction motor to be in high voltage side to achieve high efficiency. Therefore, a dc-dc converter is required to maintain the regulated constant voltage as required by the motor. Bi-directional converter is a type of converter that makes two quadrant operations of the traction motor possible. During the traction mode it acts as boost converter hence draws the power from the battery and boost it to high voltage required by the motor. During regenerative braking the same converter acts as buck converters hence utilizing the kinetic energy of the motor during braking condition and charges the battery. Regenerative braking is very useful in urban driving and helps to increase the driving range of the vehicle to some extent. Based on the operating mode of motor the closed loop system is designed to provide negated pulse as buck duty cycle negate the boost duty cycle hence power flow management is achieved.

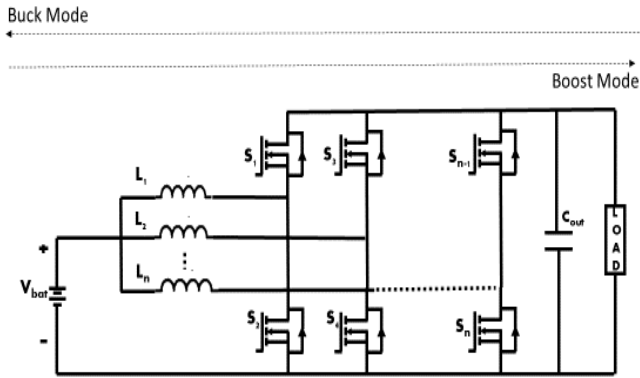


Fig.2. Multiphase bi-directional dc-dc converter

Generally, the non-isolated converters are composed of inductors, capacitors and switches and gate drivers to drive them. Losses in converters are contributed from all these components among them the dominant losses are conduction and switching losses of the switch. Conduction losses are dominant at high load and switching losses are dominant at light load condition. Inductor size is usually larger compared to other components in converters. To solve these problem multiphase converters are used. Fig.2 shows the multiphase bi-directional dc-dc converter. Multiphase converters consist are several power stages in parallel and they share common input and output capacitors. Various other advantages of multiphase converters include ripple reduction, reduced size of inductors, reduced electromagnetic emission and increased reliability etc. Multiphase bidirectional converters are employed to obtain all the above-mentioned advantages.

C. BLDC motor with electronic commutation circuit

Three-phase BLDC motor is considered in current paper. It requires reduced maintenance due to absence of brushes which makes it suitable for EV application. BLDC motor is a PMSM motor which has permanent magnets on the rotor and stator has three field windings which are placed 120° apart. Fig.3 shows the trapezoidal shape back emf produced by the motor. BLDC motor is driven by the inverter whose commutation instances are determined by three hall sensors placed on the rotor of the motor.

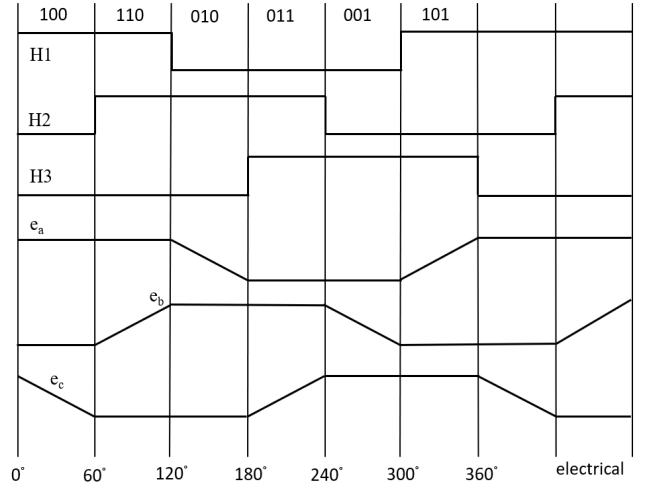


Fig.3. Hall sensor signals and back EMF waveforms [11]

III. DESIGN AND SPECIFICATIONS

The battery considered is a Li-ion battery with 24V voltage and 10Ah capacity, which drives the traction motor through boost converter. Three phase BLDC motor considered has a rating of 48V dc with rated current of 42A, rated power – 1.5kW, rated speed – 4000rpm.

The specifications for which boost converter is designed are:

- Input voltage (V_{in}) – 24V
- Output voltage (V_{out}) – 48V
- Output current (I_{out}) – 42A
- Switching frequency (F_{sw}) – 50kHz
- Output voltage ripple – 2%
- Inductor current ripple – 20%

Let the number of phases be three. Therefore, each phase is designed for $42/3 = 14A$

TABLE I. DESIGN VALUES

Parameter	Value	Unit
D (Duty ratio)	0.5	-
$I_{L(avg)}$ (per phase)	28	A
$I_{L(max)}$ (per phase)	30.8	A
$I_{L(min)}$ (per phase)	25.2	A
L_1, L_2, L_3 (Inductor per phase)	42.85	μH
C_{out} (Output capacitor)	437.5	μF

IV. SIMULATION RESULTS AND DISCUSSION

Simulation of the model is carried out using MATLAB/SIMULINK for two different cases a.) with phase shedding b.) without phase shedding. Phase shedding is a method by which based on the load current requirement the number of parallel phases in multiphase converters to be needed for operation is decided. By this at lighter loads where the switching losses is more dominant can be reduced. Similarly, as the load current increases the parallel phases could be added.

a) Without Phase shedding

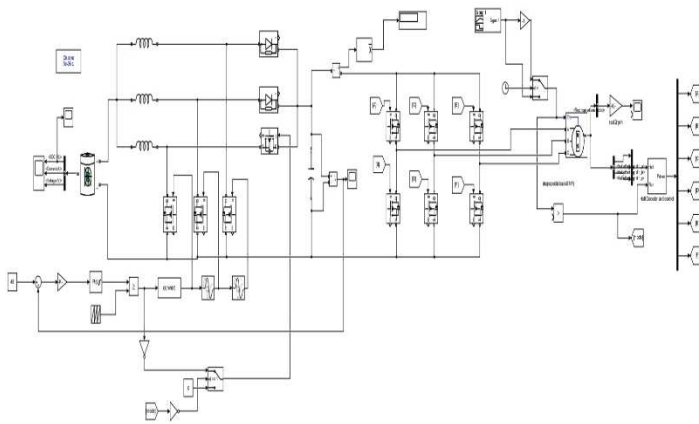


Fig.4. Simulink model without phase shedding

Fig.4 shows the simulation circuit where 24V battery is connected to the bi-directional converter which acts in a boost mode when the motor is in running condition and it acts in the buck mode when the motor is in the braking condition and simulation results are presented.

Fig.5 shows the speed variation of BLDC motor and Fig.6 shows variation in load current. Accordingly, it should be noted that till 0.43 seconds simulation time unit the torque is increased after which negative torque is provided to generate the regenerative braking condition. During which current is negative implying that its direction is reversed and the SOC of the battery is increasing indicating that the battery is getting recharged as shown in Fig. 7

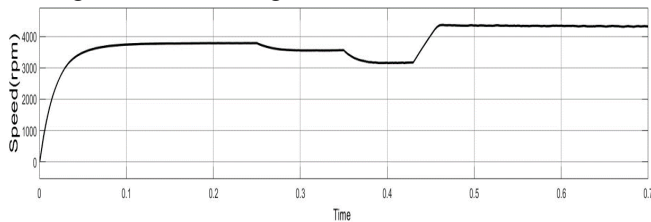


Fig.5. Variation of speed of BLDC motor with variation in loads over a period

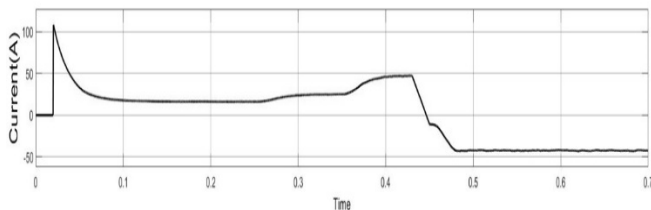


Fig.6. Current drawn by BLDC motor during various load conditions

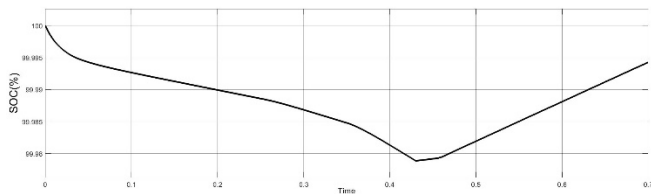


Fig7. SOC of a battery during motoring and regenerative braking mode

b) With Phase shedding algorithm

Fig. 8 shows the simulation circuit with same parameters as in earlier case but with phase shedding implemented. Phase shedding algorithm is implemented using a MATLAB function block which checks the load current and a delay time is given till the steady state is reached after which the

phase shedding algorithm will be implemented, Fig.9 gives the flowchart.

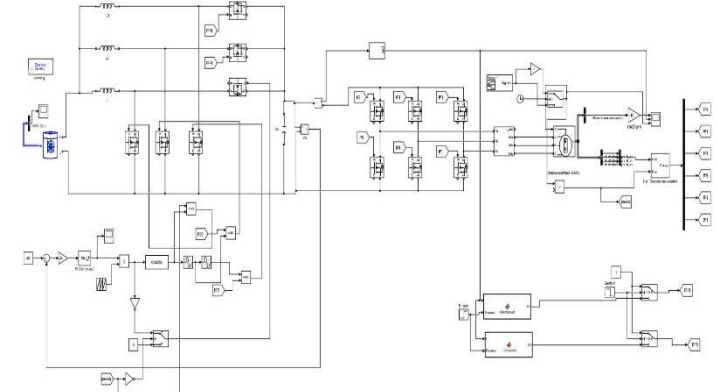


Fig.8. Simulink model with phase shedding

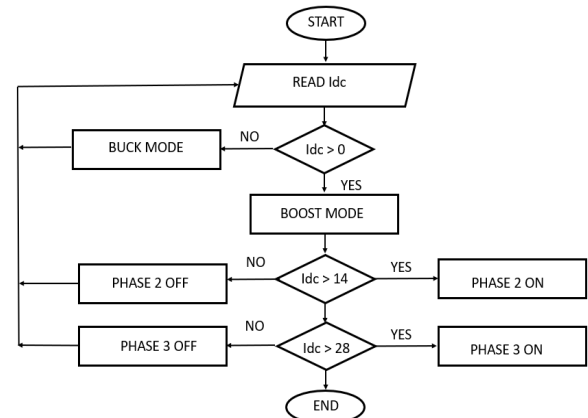


Fig.9. Flowchart of Phase shedding

As mentioned in the design earlier each phase is capable of handling 14A of current hence as the load varies the phases are shed and added accordingly. Simulation results show that the performance of the system is not affected by phase shedding Fig 10.a shows speed of traction motor and Fig.10.b show variation in load current and Fig.10.c shows varying torque condition. Regenerative braking scenario during which battery is getting recharged is as shown in Fig.11 where the SOC of the battery increases. As all the switches are not operated all time switching losses will be reduced aiding to improved efficiency especially at light load condition.

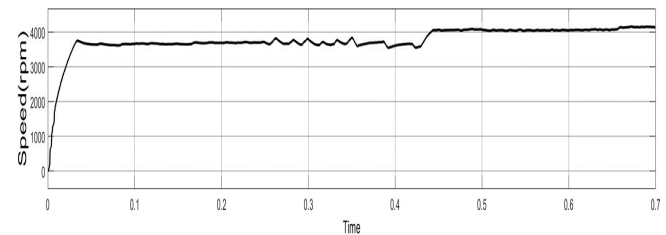


Fig.10.a Variation of speed of BLDC motor with variation in loads over a period

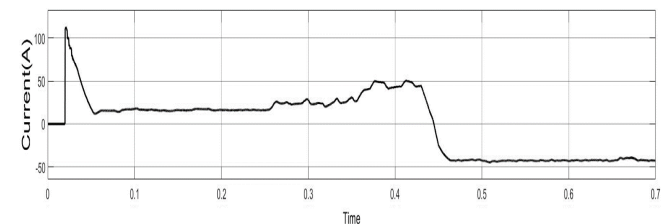


Fig.10.b Current drawn by BLDC motor during various load conditions

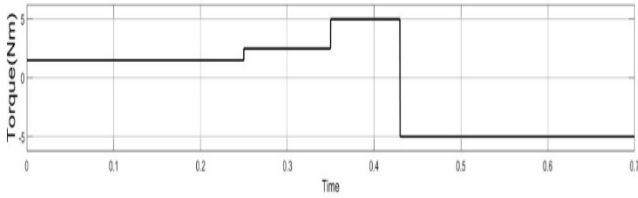


Fig.10.c Variation in torque over period

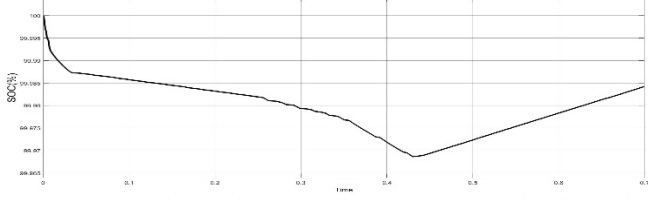


Fig.11. SOC of a battery during motoring and regenerative braking mode

V. SOFT SWITCHING

A. Soft switching topology

Multiphase converters offer several benefits as discussed earlier, however there is an increased number of switches involved in the process compared to single phase. By employing phase shedding method power loss in the converter section could be reduced hence efficiency can be improved. Another method of reducing the power loss is by soft switching method. Soft switching is a method of turning on or off a switch at an instance when either voltage or current across the switch is zero hence reducing the switching loss. Fig. 12 shows hard switching MOSFET and power loss regions.

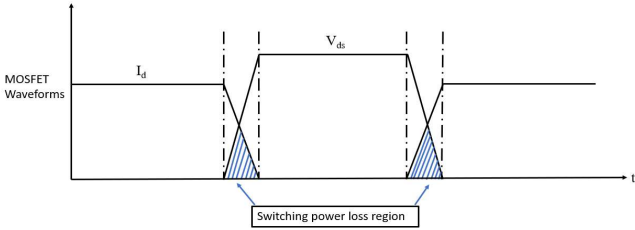


Fig.12. Hard switching MOSFET current and voltage waveforms

Fig. 13 shows soft switching boost converter topology used in current paper. It consists of resonant inductor L_r , Auxiliary capacitor C_a , and resonant capacitor C_r and two diodes D_1 and D_2 [12].

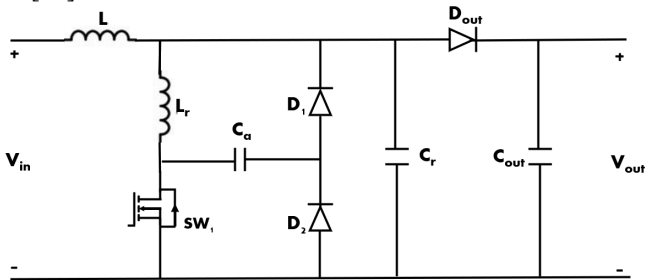


Fig.13. Soft switching boost converter topology [12]

Assumptions made for explaining the modes of operation are:

- 1) The passive components and switches are assumed to be ideal and lossless
- 2) Initial conditions of each operating mode are assumed to be zero
- 3) The parasitic component of switches and other components are neglected

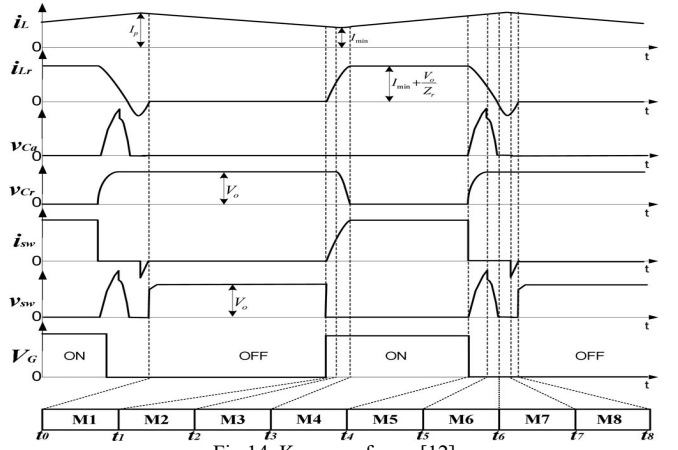


Fig.14. Key waveforms [12]

Fig. 14 gives the key waveforms with eight operating modes with each mode details presented below.

Mode 1 ($t_0 \leq t < t_1$): During this mode, main inductor charged in the previous cycle starts to discharge through diode D_{out} to load. Inductor current is decreasing, and inductor voltage becomes $\{-(V_0 - V_{in})\}$.

Mode 2 ($t_1 \leq t < t_2$): Switch is closed at zero current (ZCS) due to Resonant inductor (L_r), the current i_{Lr} increases linearly as output voltage is supplied to L_r . The diode current D_{out} reaches zero when i_{Lr} is equal to i_L .

Mode 3 ($t_2 \leq t < t_3$): Current through inductor $L(i_L)$ flows through resonant inductor and switch reaching its lower threshold value. The current i_{D_o} reaches zero. L_r and C_r resonate and the voltage across resonant capacitor V_{Cr} drops from output voltage V_o to zero.

Mode 4 ($t_3 \leq t < t_4$): The voltage across resonant capacitor V_{Cr} is zero and diodes D_1 and D_2 starts to conduct, i_L current starts to increase linearly during this period. The resonant inductor current i_{Lr} has two parts, main inductor current i_L and current through D_1 and D_2 .

Mode 5 ($t_4 \leq t < t_5$): Switch is open at zero voltage due to the auxiliary capacitor C_a . There are two current paths. One is L - C_r - V_{in} , voltage across the resonant capacitor V_{Cr} increases from zero to output voltage V_o . The other path is L_r - C_a - D_1 where the second resonance occurs, during which stored energy in inductor L_r is transferred to C_a due to which i_{Lr} decreases and V_{ca} reaches maximum.

Mode 6 ($t_5 \leq t < t_6$): This mode begins when V_{Cr} equals V_o . Energy transfer takes place from L_r to C_a and i_{Lr} reduces to zero.

Mode 7 ($t_6 \leq t < t_7$): In this mode resonance occurs in path D_2 - C_a - L_r - D_{out} - C_o and energy transfers from C_a to L_r . V_{ca} decreases and i_{Lr} is in the reverse direction when V_{ca} reaches zero antiparallel diode of switch turns on.

Mode 8 ($t_7 \leq t < t_8$): Inductor L supplies output through D_{out} hence i_L decreases. Resonant inductor i_{Lr} also transfers energy to the load through D_{out} , this mode is completed when i_{Lr} becomes zero.

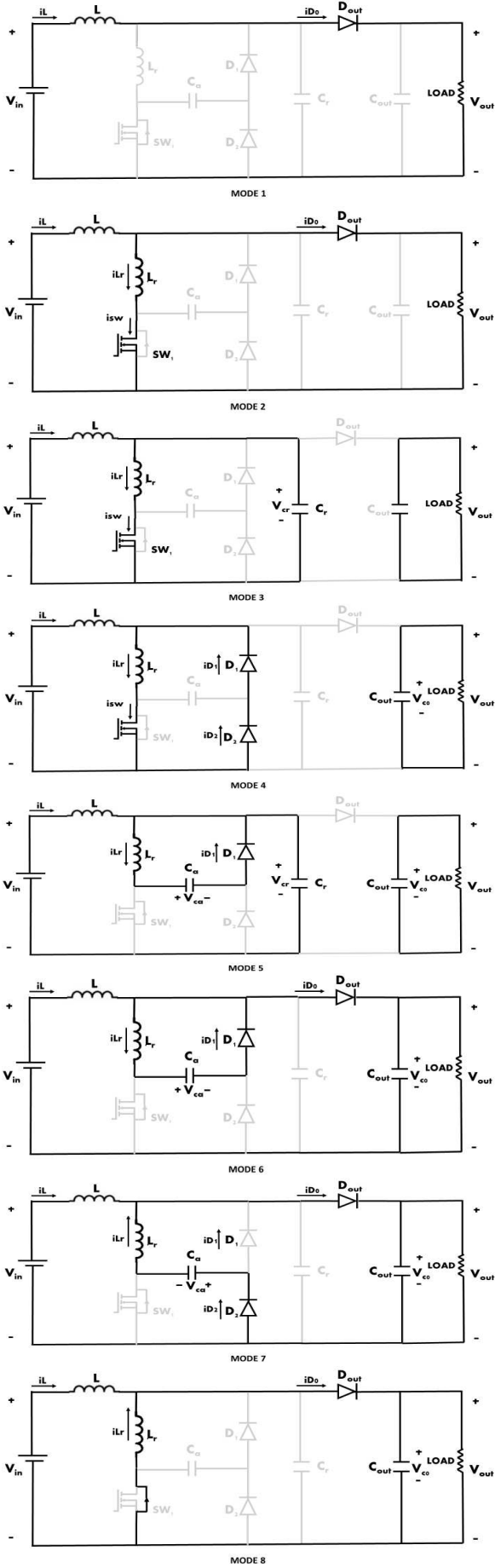


Fig.15. Operational modes of resonant converter [12]

B. Selection of resonant inductor and capacitor

In mode 5 the current that flows through the path $L_r - C_a - D_1$ should be large enough for resonance. The voltage across the switch during this mode is given by expression

$$V_{sw}(t) = V_{ca}(t) + V_{cr}(t) \quad (1)$$

The auxiliary capacitance C_a must be 20 times more than the output capacitance of switch, as C_a is charged by the resonant inductor current during switch turn-off period.

$$C_a > 20C_{oes} \quad (2)$$

Duty ratio of boost converter is given by

$$D = 1 - (V_{in}/V_{out}) \quad (3)$$

The resonant inductor L_r is given by expression

$$L_r = \{V_0 / (I_2 - I_{min})\}^2 C_r \quad (4)$$

$$\text{where } I_2 - I_{min} = V_0 / Z_r = V_0 (\sqrt{C_r} / \sqrt{L_r}) \quad (5)$$

The resonant capacitor is given by the expression

$$C_r = (D_{min}^2 / \Pi^2 L_{sw}^2) + (I_{min}^2 L_r / \Pi^2 V_0^2) \quad (6)$$

TABLE II. SIMULATION PARAMETERS FOR INTERLEAVED SOFT SWITCHING

Parameter	Symbol	Value	Unit
Input voltage	V_{in}	24	V
Output voltage	V_{out}	48	V
Rated Power	P_0	2	kW
Main Inductor per phase	L_1, L_2, L_3	42.85	μH
Resonant inductor	L_{r1}, L_{r2}, L_{r3}	2	μH
Resonant capacitor	C_{r1}, C_{r2}, C_{r3}	1.5	μF
Auxiliary capacitor	C_{a1}, C_{a2}, C_{a3}	200	nF
Output capacitor	C_{out}	437.5	μF
Switching frequency	F_{sw}	50	kHz

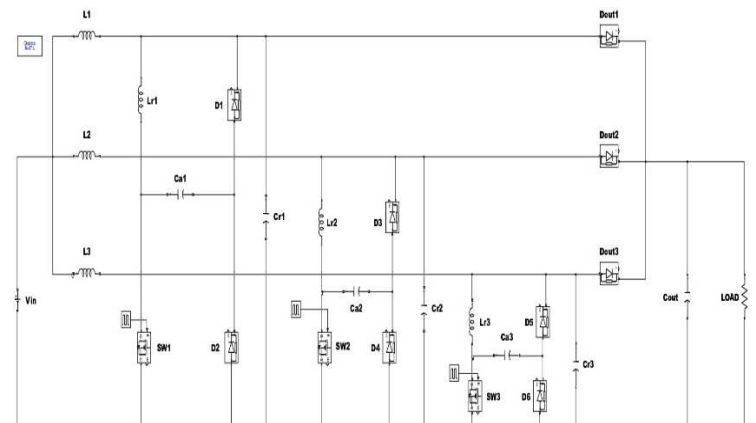


Fig.16. Simulation circuit for three phase interleaved soft switching boost converter

Fig. 16 shows the simulation circuit for soft switching of three phase boost converter circuit, the simulation parameters used are listed in Table 2.

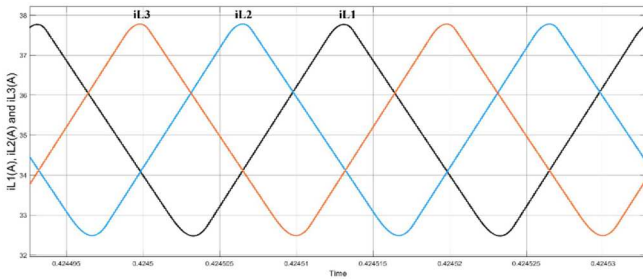


Fig. 17. Current through main inductor L1, L2 and L3

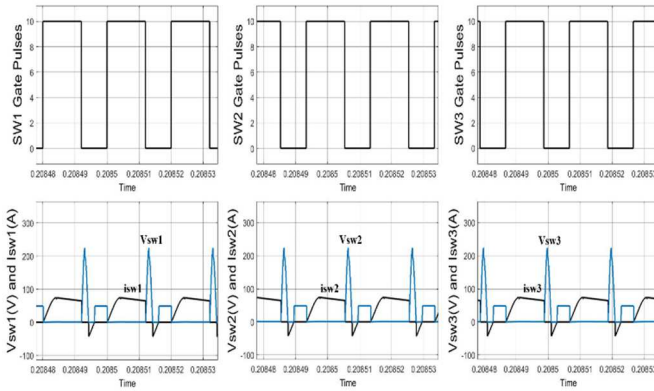


Fig. 18. Waveform of SW1, SW2 and SW3 Gate Pulses, and Current and voltage across SW1, SW2 and SW3

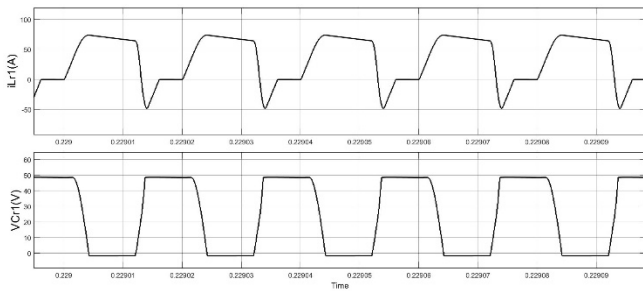


Fig. 19.a. Waveform of current through resonant inductor L_{r1} ,
Fig. 19.b. Voltage across resonant capacitor C_{r1}

Fig. 17 shows the current through the inductors L1, L2 and L3. Fig. 18 shows voltage and current waveforms of the switches and gate pulse signals. The switches SW1 and SW2 and SW3 are turned on under ZCS and turned off under ZVS achieving soft switching. Fig. 19 shows the current and voltage across resonant inductor and capacitor respectively.

VI. CONCLUSION

A battery-powered electric vehicle driven by BLDC motor in which motor is operating in motoring mode and braking mode is considered. With regenerative braking the kinetic energy of the vehicle during braking can be utilized and stored in battery, where it can be used later to drive the vehicle. Multiphase bi-directional converters are used, and the phase shedding technique is employed. The selection of number of active phases as per the load requirement is known as phase shedding. In this project a topology of multiphase bidirectional dc-dc converter with phase shedding algorithm is proposed for BLDC driven Electric Vehicle (EV) application. The number of phases is determined based on the

current drawn by motor and simulation work is carried out using MATLAB/Simulink. Soft switching technique is employed for three phase interleaved boost converter for reducing the switching losses and simulation results are presented. In the future hardware prototype will be made and simulation results will be compared.

REFERENCES

- [1] K. Poornesh, K. P. Nivya and K. Sireesha, "A Comparative study on Electric Vehicle and Internal Combustion Engine Vehicles," 2020 International Conference on Smart Electronics and Communication (ICOSEC), Trichy, India, 2020, pp. 1179-1183, doi: 10.1109/ICOSEC49089.2020.9215386.
- [2] S. S. Bhurse and A. A. Bhole, "A Review of Regenerative Braking in Electric Vehicles," 2018 International Conference on Computation of Power, Energy, Information and Communication (ICCPEIC), Chennai, 2018, pp. 363-367, doi: 10.1109/ICCPEIC.2018.8525157.
- [3] A. T. Waghe and S. K. Patil, "A bidirectional DC-DC converter fed separately excited DC motor electric vehicle application," 2020 International Conference on Emerging Trends in Information Technology and Engineering (ic-ETITE), Vellore, India, 2020, pp. 1-5, doi: 10.1109/ic-ETITE47903.2020.421.
- [4] A. Sah, K. Chaudhary and V. V. Ratnam, "Non-isolated multiphase buck-boost converter design for electric vehicle applications," 2014 Annual International Conference on Emerging Research Areas: Magnetics, Machines and Drives (AICERA/iCMMD), Kottayam, 2014, pp. 1-6, doi: 10.1109/AICERA.2014.6908181.
- [5] A. M. Youssef, "Multiphase Interleaved Bidirectional DC/DC Converter for Fuel Cell/Battery Powered Electric Vehicles," 2020 12th International Conference on Electrical Engineering (ICEENG), Cairo, Egypt, 2020, pp. 34-43, doi: 10.1109/ICEENG45378.2020.9171753.
- [6] M. P. Varghese, A. Manjunatha and S. Kumar, "Performance analysis of multiphase buck converter for VRMs using phase shedding technique," 2016 Biennial International Conference on Power and Energy Systems: Towards Sustainable Energy (PESTSE), Bangalore, 2016, pp. 1-5, doi: 10.1109/PESTSE.2016.7516529.
- [7] M. Marcinek and M. Holub, "Multiphase, Synchronous GaN Buck Converters - Efficiency Based Selection of the Number of Phases," 2018 20th European Conference on Power Electronics and Applications (EPE'18 ECCE Europe), Riga, 2018, pp. P.1-P.9.
- [8] N. Hashemnia and B. Asaei, "Comparative study of using different electric motors in the electric vehicles," 2008 18th International Conference on Electrical Machines, Vilamoura, Portugal, 2008, pp. 1-5, doi: 10.1109/ICELMACH.2008.4800157.
- [9] S. K. Mazumder, K. Acharya and P. Jedraszczak, "High-temperature all-SiC bidirectional DC/DC converter for plug-in-hybrid vehicle (PHEV)," 2008 34th Annual Conference of IEEE Industrial Electronics, Orlando, FL, USA, 2008, pp. 2873-2878, doi: 10.1109/IECON.2008.4758415.
- [10] A. K. M. Ahasan Habib, S. M. A. Motakabber and M. I. Ibrahimy, "A Comparative Study of Electrochemical Battery for Electric Vehicles Applications," 2019 IEEE International Conference on Power, Electrical, and Electronics and Industrial Applications (PEEIACON), Dhaka, Bangladesh, 2019, pp. 43-47, doi: 10.1109/PEEIACON48840.2019.9071955.
- [11] Swathi Kumari, Y. Laxminarayana, Dr. S. Tarakalyani, "Modeling and Simulation of Bldc Motor for Aiding and Opposing Loads," 2013 IOSR Journal of Electrical and Electronics Engineering (IOSR-JEEE), e-ISSN: 2278-1676, p-ISSN: 2320-3331, Volume 7, Issue 4
- [12] D. Jung, Y. Ji, S. Park, Y. Jung and C. Won, "Interleaved Soft-Switching Boost Converter for Photovoltaic Power-Generation System," in IEEE Transactions on Power Electronics, vol. 26, no. 4, pp. 1137-1145, April 2011, doi: 10.1109/TPEL.2010.2090948.